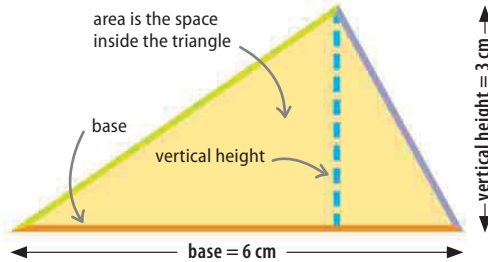


Finding the area of a triangle

To calculate the area of a triangle, substitute the given values for the base and vertical height into the formula. Then work through the multiplication shown by the formula ($\frac{1}{2} \times \text{base} \times \text{vertical height}$).

▷ An acute-angled triangle

The base of this triangle is 6 cm and its vertical height is 3 cm. Find the area of the triangle using the formula.



First, write down the formula for the area of a triangle.

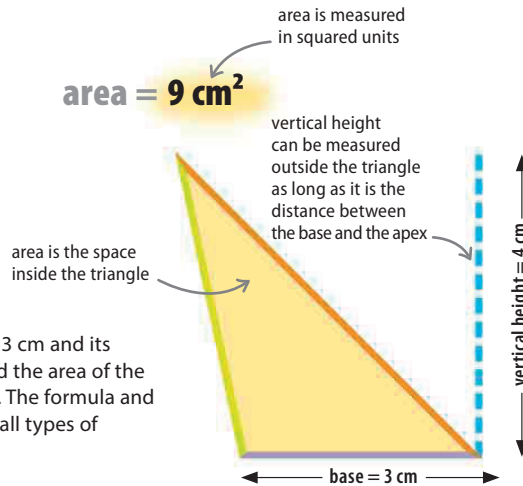
$$\text{area} = \frac{1}{2} \times \text{base} \times \text{vertical height}$$

Then, substitute the lengths that are known into the formula.

$$\text{area} = \frac{1}{2} \times 6 \times 3$$

Work through the multiplication in the formula to find the answer. In this example, $\frac{1}{2} \times 6 \times 3 = 9$. Add the units of area to the answer, here cm^2 .

$$\text{area} = 9 \text{ cm}^2$$



First, write down the formula for the area of a triangle.

$$\text{area} = \frac{1}{2} \times \text{base} \times \text{vertical height}$$

Then, substitute the lengths that are known into the formula.

$$\text{area} = \frac{1}{2} \times 3 \times 4$$

Work through the multiplication to find the answer, and add the appropriate units of area.

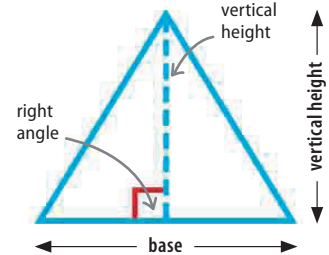
$$\frac{1}{2} \times 3 \times 4 = 6$$

$$\text{area} = 6 \text{ cm}^2$$

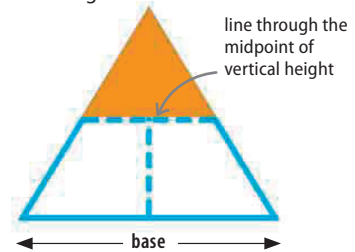
LOOKING CLOSER

Why the formula works

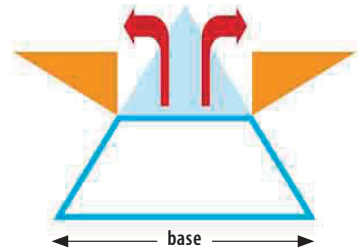
By adjusting the shape of a triangle, it can be converted into a rectangle. This process makes the formula for a triangle easier to understand.



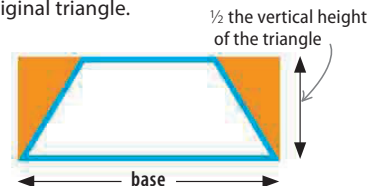
Draw any triangle and label its base and vertical height.



Draw a line through the midpoint of the vertical height that is parallel to the base.



This creates two new triangles. These can be rotated around the triangle to form a rectangle. This has exactly the same area as the original triangle.



The original triangle's area is found using the formula for the area of a rectangle ($b \times h$). Both shapes have the same base; the rectangle's height is $\frac{1}{2}$ the height of the triangle. This gives the area of the triangle formula: $\frac{1}{2} \times \text{base} \times \text{vertical height}$.